

SECTION 26 05 19 - CONDUCTORS

PART 1 – GENERAL

1.01 SECTION INCLUDES

- A. Conductors for power, lighting, sound, communication and control, including:
 - 1. Conductors for general wiring.
 - 2. Multi-conductor cable.
 - 3. Flexible cords and cables.

1.02 RELATED WORK SPECIFIED ELSEWHERE

- A. Conductor identification: Section 26 05 53.
- B. Conductor connections and devices: Section 26 05 21.
- C. Wiring Devices: Section 26 27 26.
- D. Grounding: Section 26 05 26.

1.03 REFERENCE SPECIFICATIONS AND STANDARDS

- A. National Electrical Code. Reference Articles 250, 310, and 800.

PART 2 – PRODUCTS

2.01 ACCEPTABLE MANUFACTURERS

- A. Cable products as manufactured by the following manufacturers only are acceptable in meeting specified requirements:
 - 1. General Cable Co.
 - 2. Rome Cable.
 - 3. American Insulated Wire.
 - 4. BICC Cable Corp.
 - 5. 5. Belden Inc.

2.02 MATERIAL AND FABRICATION

- A. Conductors for general wiring: Thermoplastic insulation rated for 600V manufactured in accord with UL 83.
 - 1. Copper conductors, 3/4 hard drawn. Type THHN/THWN or XHHN/XHHW where required by location. Markings on cables shall be clearly marked with size and

type. Markings shall be white lettering with black jacketing for conductor sizes of #6 AWG or larger. Indentations for lettering are not acceptable. Cable sizes #8 AWG and larger shall be stranded conductors.

PART 3 – EXECUTION

3.01 USE

A. Conductors for general wiring:

1. Minimum 90°C temperature rated insulation on conductors, except minimum 105°C temperature rated insulation on conductors in conduits exposed on roof.
2. Stranded conductors at motors and other applications where subject to vibration.
3. Minimum size conductors for power and lighting: #12 AWG, with THHN/THWN insulation except where noted.
4. Minimum size conductors for motor control circuits: #14 AWG stranded with THHN/THWN insulation except where noted.
5. All cable conductors of 3-phase circuits or single-phase shall be of the same type, mixing stranded conductors for some phases with others as solid is not acceptable.
6. Multi-conductor cable: Only for use in low voltage control circuits.

B. Ground conductors:

1. Bare copper conductor may be used for equipment ground only.
 - a. Install ground conductors in all feeder conduits and all non-metallic conduits as required by Electrical Code and Section 26 05 26.
2. Insulated ground conductors:
 - a. Ground conductor insulation color: Green.
 - b. Isolated ground conductor insulation color: Green with white stripe.

3.02 INSPECTION

- A. Check conduit system for damage and loose connections, replace damaged sections.
- B. Check for caps at conduit openings. Make sure that inside of conduit is free of dirt and moisture.
- C. Pull mandrel, one size smaller than the conduit, through entire length of all underground conduits prior to conductor installation.

3.03 INSTALLATION

A. Conductors for general wiring:

1. Color code conductors insulation as follows:

SYSTEM VOLTAGE		
Conductor	208Y/120	480Y/277
Phase A	Black	Brown
Phase B	Red	Orange
Phase C	Blue	Yellow

2. For conductors #6 AWG or larger, permanent plastic colored tape may be used to mark conductor end in lieu of coded insulation. Tape shall cover not less than 2 in. of conductor insulation within enclosure.
3. When pulling conductors do not exceed manufacturer's recommended values.
4. Use polypropylene or nylon ropes for pulling conductors.

B. Prior to energizing, ensure that cables are:

2. Free from short circuits and grounds.
3. Free from mechanical and electrical defects.

C. Test all conductors in accordance with Section 26 05 53.

D. Conductor identification:

1. Feeders: Identify with corresponding circuit designation at over-current device and load ends, at all splices, and in pull boxes.
2. Provide labels as indicated in Section 26 05 53, and on plans and details.

E. Multi-conductor and twin axial cables: Install cables end-to-end without splices.

END OF SECTION 26 05 19